Experiment Title

Write a Program in Python for Principal Component Analysis on Smartphone Dataset

Aim

To implement Principal Component Analysis (PCA) on a Smartphone Dataset using Python and reduce the dimensionality of the dataset while preserving maximum variance present in the original data.

Objectives

Upon successful completion of this experiment, students will be able to:

1. Understand the concept of dimensionality reduction.
2. Apply Principal Component Analysis (PCA) on real-world data.
3. Select and preprocess numerical features for PCA.
4. Handle missing values in a dataset.
5. Standardize data before applying PCA.
6. Analyze explained variance and cumulative explained variance.
7. Reduce the number of features without significant loss of information.
8. Visualize data using principal components.
9. Interpret PCA results for decision-making.

Theory

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is an unsupervised machine learning technique used for dimensionality reduction and feature extraction. It transforms a large set of correlated variables into a smaller set of uncorrelated variables known as **Principal Components (PCs)**.

The first principal component captures the maximum variance present in the dataset. Each succeeding principal component captures the remaining variance while maintaining orthogonality with previous components.

PCA helps in:

- Reducing computational complexity.
- Eliminating redundant information.
- Improving data visualization.
- Reducing noise in the dataset.

- Enhancing model performance.

Key Concepts of PCA

- **Principal Component (PC):** A new feature formed as a linear combination of original features.
- **Explained Variance:** The amount of information captured by a principal component.
- **Cumulative Variance:** The total variance retained by multiple principal components.
- **Feature Scaling:** Standardization is required because PCA is sensitive to feature magnitudes.

Applications of PCA

- Image Compression
- Face Recognition
- Recommendation Systems
- Bioinformatics
- Financial Analysis
- Smartphone Specification Analysis
- Market Research

Dataset Description

The experiment uses the dataset: smartphone_cleaned_data.csv

The dataset contains various smartphone-related attributes such as:

- Price
- Rating
- Display Specifications
- Camera Specifications
- RAM
- Storage
- Battery Capacity
- Processor Features
- Other Smartphone Characteristics

The dataset is analyzed to identify the most important underlying features contributing to variations among smartphones.

Algorithm

Principal Component Analysis on Smartphone Dataset

1. Import required Python libraries.
2. Load the smartphone dataset.
3. Display basic dataset information.
4. Generate statistical summary of the dataset.
5. Check for missing values and duplicate records.
6. Visualize correlation among numerical features.
7. Select numerical features from the dataset.
8. Handle missing values using median imputation.
9. Standardize the numerical data.
10. Apply Principal Component Analysis (PCA).
11. Calculate explained variance ratio.
12. Calculate cumulative explained variance.
13. Reduce data to two principal components.
14. Visualize the transformed dataset using a scatter plot.
15. Interpret the PCA results.

Program:

```
# Principal Component Analysis on Smartphone Dataset
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
# Display plots inside Jupyter Notebook
%matplotlib inline
# Set plotting style
sns.set(style="whitegrid")
# Load Dataset
```

```
df = pd.read_csv("smartphone_cleaned_data.csv")
print("Dataset Shape:", df.shape)
# Select Numerical Features
numeric_df = df.select_dtypes(include=np.number)
# Handle Missing Values
for column in numeric_df.columns:
    numeric_df[column] = numeric_df[column].fillna(
        numeric_df[column].median()
    )
# Standardize Data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(numeric_df)
print("Shape of Standardized Data:")
print(X_scaled.shape)
# Apply PCA
pca = PCA()
X_pca = pca.fit_transform(X_scaled)
print("\nOriginal Shape:", X_scaled.shape)
print("PCA Shape:", X_pca.shape)
# Explained Variance Ratio
print("\nExplained Variance Ratio")
for i, variance in enumerate(
        pca.explained_variance_ratio_,
        start=1):
    print(f"PC{i}: {variance*100:.2f}%")
# Cumulative Explained Variance
cumulative_variance = np.cumsum(
    pca.explained_variance_ratio_
)
print("\nCumulative Explained Variance")
for i, variance in enumerate(
        cumulative_variance,
```

```
        start=1):
    print(f"PC1 to PC{i}: {variance*100:.2f}%")
# Reduce Data to Two Principal Components
pca_2 = PCA(n_components=2)
X_pca_2 = pca_2.fit_transform(X_scaled)
print(
    "\nTotal Variance Explained by First Two Components:"
)
print(
    np.sum(
        pca_2.explained_variance_ratio_
    ) * 100
)
# Create DataFrame of Principal Components
pca_df = pd.DataFrame(
    X_pca_2,
    columns=["PC1", "PC2"]
)
print("\nFirst Five Records")
print(pca_df.head())
# Visualize PCA Components
plt.figure(figsize=(8,6))
plt.scatter(
    pca_df["PC1"],
    pca_df["PC2"],
    alpha=0.7
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title(
    "PCA on Smartphone Dataset"
)
```

```
plt.grid(True)
plt.show()
```

Sample Output

```
Dataset Shape: (980, 14)
Shape of Standardized Data:
(980, 14)
Original Shape: (980, 14)
PCA Shape: (980, 14)
Explained Variance Ratio
PC1: 35.42%
PC2: 22.18%
PC3: 11.35%
PC4: 8.23%
...
Cumulative Explained Variance
PC1 to PC1: 35.42%
PC1 to PC2: 57.60%
PC1 to PC3: 68.95%
PC1 to PC4: 77.18%
...
Total Variance Explained by First Two Components:
57.60
First Five Records
       PC1      PC2
0   1.2454  -0.4532
1   0.9856   0.7482
2  -1.2568   1.3245
3   2.1478  -0.8927
4  -0.6543  -1.1235
```

Observations

1. The smartphone dataset was successfully loaded and processed.
2. Numerical features were selected for PCA analysis.
3. Missing values were handled using median imputation.
4. Data was standardized before PCA execution.
5. PCA transformed the original features into principal components.
6. The first few principal components captured a significant amount of total variance.
7. Dimensionality reduction was successfully achieved.
8. The transformed dataset was visualized using the first two principal components.

Result

Thus, Principal Component Analysis (PCA) was successfully implemented on the Smartphone Dataset using Python. The dataset was standardized and transformed into principal components, reducing dimensionality while preserving maximum variance. The transformed data was visualized using the first two principal components for analysis and interpretation.

Conclusion

Principal Component Analysis is a powerful dimensionality reduction technique that transforms high-dimensional smartphone data into a smaller set of informative principal components. The technique reduces feature redundancy, improves data visualization, and preserves most of the valuable information contained in the original dataset, making it an essential preprocessing step in many machine learning applications.

Viva Voce Questions

1. What is Principal Component Analysis (PCA)?
2. Why is PCA used in Machine Learning?
3. What is dimensionality reduction?
4. What are principal components?
5. Why is standardization necessary before applying PCA?
6. What is the role of StandardScaler in PCA?
7. What is explained variance ratio?
8. What is cumulative explained variance?
9. What are eigenvalues and eigenvectors?

10. Why are only numerical features used in PCA?
11. What is feature extraction?
12. What is the difference between feature selection and feature extraction?
13. How do we decide the number of principal components?
14. What are the advantages of PCA?
15. What are the limitations of PCA?
16. Is PCA a supervised or unsupervised learning technique?
17. Can PCA be used for visualization? Explain.
18. What happens if PCA is applied without feature scaling?
19. How does PCA help in reducing computational complexity?
20. Mention some real-world applications of PCA.